

Unit 1: Ratios and Proportions

Topic 1: Data Analysis — Challenge Level 2 🍌🍌

Name: _____

Date: _____

- Q1.** Shop A sells 3 apples for \$2. Shop B sells 5 apples for \$3. Which is cheaper?
- (A) Shop A
(B) Shop B
(C) Same
(D) Not enough info
- Q2.** A double number line shows 0 to 10 miles on top and 0 to 15 minutes on bottom. What is the value for 4 miles?
- (A) 5 min
(B) 6 min
(C) 9 min
(D) 12 min
- Q3.** The point (4, 10) is on a ratio graph. Which of these points is also on the same line?
- (A) (2, 5)
(B) (6, 12)
(C) (8, 15)
(D) (1, 3)
- Q4.** Ratio of $A : B$ is 3:4. Ratio of $B : C$ is 4:5. What is the ratio of $A : C$?
- (A) 3:5
(B) 4:3
(C) 5:3
(D) 1:1
- Q5.** A 250ml drink has 40g of sugar. How much sugar is in 1 liter (1000ml)?
- (A) 80g
(B) 120g
(C) 160g
(D) 200g
- Q6.** On a map, 3cm = 150km. What is the distance between two cities 7cm apart?
- (A) 300km
(B) 350km
(C) 450km
(D) 700km
- Q7.** A student reads 12 pages in 20 minutes. How many pages can they read in 1 hour?
- (A) 24
(B) 30
(C) 36
(D) 40
- Q8. [FRQ]** Use a table to compare: Brand X costs \$12 for 3 lbs. Brand Y costs \$18 for 4 lbs. Show which brand is the better buy.
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- Q9. [FRQ]** Plot the following ratio pairs on a coordinate plane (mentally or describe): (1, 2), (2, 4), (3, 6). Is the relationship proportional? Why?
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- Q10. [FRQ]** The ratio of wings to beaks in a zoo house is 2:1. If 5 more birds fly in, and the new total count of beaks is 15, how many wings are there now?
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💡 Chef's Tips

- **Unit Pricing:** Find the cost of "1 unit" to compare different shops.
- **Double Number Lines:** Keep the top and bottom values aligned to maintain the ratio.
- **Graphing:** A proportional ratio graph always forms a straight line starting from (0,0).